

№11

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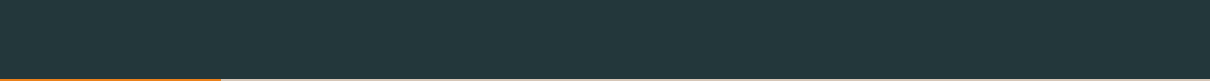


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- 1032225355@pfur.ru
- <https://nvemangue.github.io/ru/>

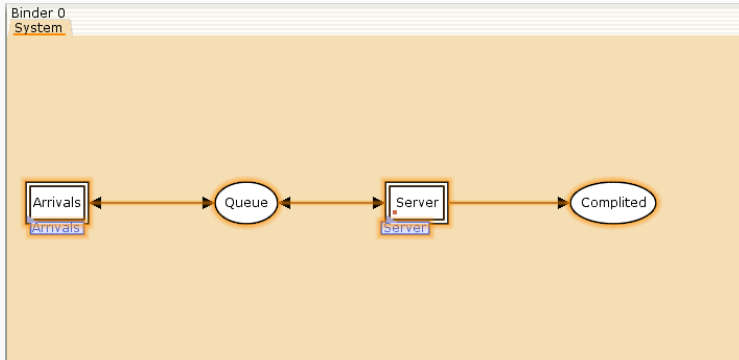


$M|M|1$ CPN tools.

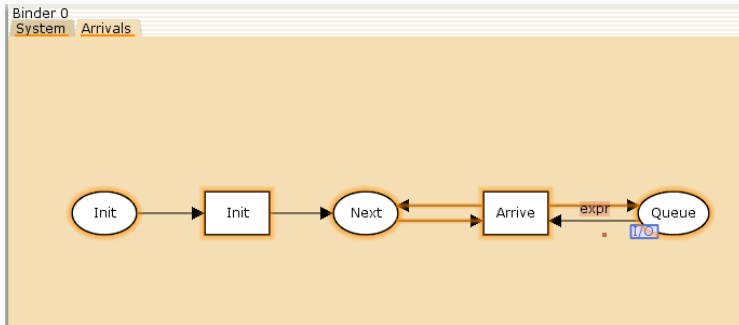
- CPN Tools
- $M|M|1$.



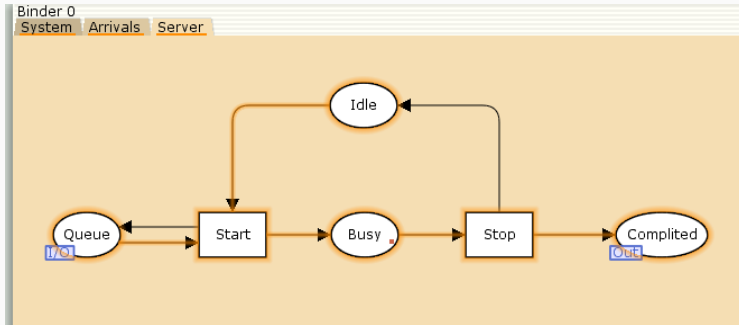
... - FIFO. (),



. 1:



. 2:



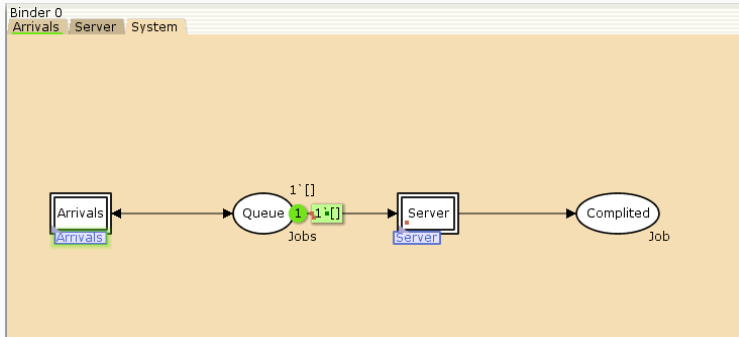
. 3:


```

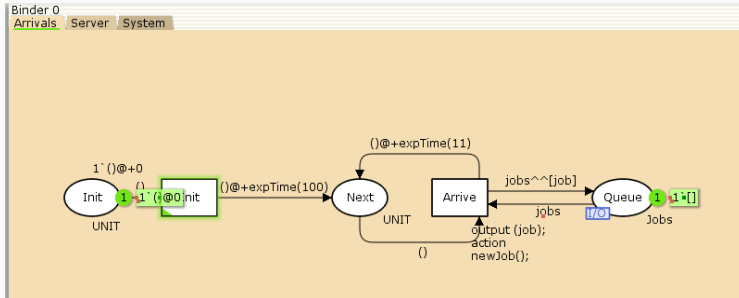
▼ Declarations
  ▼ Standard declarations
    ► colset BOOL
    ► colset STRING
  ▼ System
    ▼ colset UNIT = unit timed;
    ▼ colset INT = int;
    ▼ colset Server = with server timed;
    ▼ colset JobType = with A|B;
    ▼ colset Job = record
      jobType : JobType * AT : INT;
    ▼ colset Jobs = list Job;
    ► colset ServerxJob
    ▼ var proctime : INT;
    ▼ var job : Job;
    ▼ var jobs : Jobs;
    ▼ fun expTime (mean : int) =
      let
        val realMean = Real.fromInt mean
        val rv = exponential ((1.0/realMean))
      in
        floor (rv+0.5)
      end;
    ▼ fun intTime() = IntInf.toInt (time());
    ▼ fun newJob() = {jobType = JobType.ran(), AT = intTime() };
▼ Monitors

```

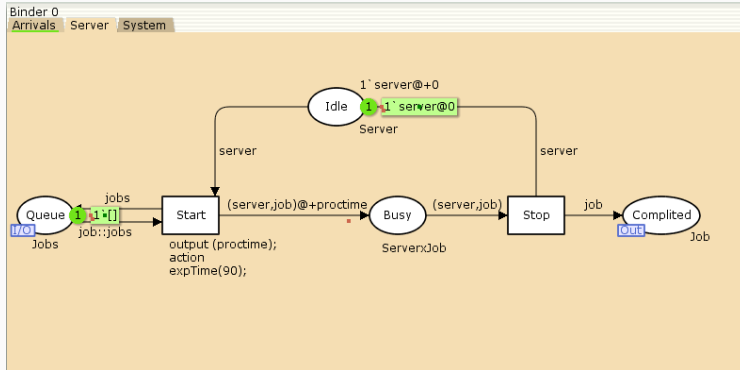




. 5:



. 6:



. 7:

```

Binder 0
Arrivals System Server fun pred <Ostanovka>
fun pred (bindelem) =
let
  fun predBindElem (Server'Start (1,
                                {job,jobs,proctime})) = Queue_Delay.count()=200
  | predBindElem _ = false
in
  predBindElem bindelem
end

```

. 8: Predicate Ostanovka

```

Binder 0
Arrivals System Server fun obs<Queue Delay>
fun obs (bindelem) =
let
  fun obsBindElem (Server'Start (1, {job,jobs,proctime})) =
(intTime() - (#AT job))|
    | obsBindElem _ = ~1
in
  obsBindElem bindelem
end

```

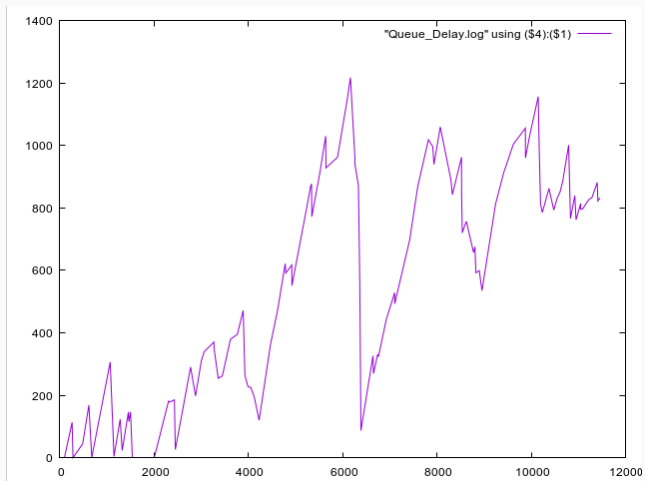
. 9: Observer Queue Delay

```

/home/openmodelica/output/logfiles/Queue_Delay.log - Mousepad
Файл  Правка  Поиск  Вид  Документ  Справка
#data counter step time
0 1 3 496
0 2 6 507
0 3 9 523
97 4 22 624
134 5 27 662
266 6 45 808
308 7 50 850
356 8 57 913
347 9 60 915
341 10 63 920
501 11 78 1085
664 12 97 1256
|

```

. 10: Queue_Delay.log



. 11:

Binder 0
Arrivals System Server fun obs <Queue Delay Real>

```
fun obs (bindelem) =  
  let  
    fun obsBindElem (Server'Start (1, {job,jobs,proctime})) =  
      Real.fromInt(intTime()-(#AT job))  
      | obsBindElem _ = ~1.0  
  in  
    obsBindElem bindelem  
  end
```

. 12: Observer Queue Delay Real

```

/home/openmodelica/output/logfiles/Queue_Delay_Real.log - M
Файл  Правка  Поиск  Вид  Документ  Справка
#data counter step time
0.000000 1 3 69
60.000000 2 6 197
135.000000 3 9 373
75.000000 4 12 499
0.000000 5 15 777
12.000000 6 18 844
0.000000 7 21 957
175.000000 8 27 1167
397.000000 9 30 1419
459.000000 10 34 1537
647.000000 11 39 1726
536.000000 12 41 1879
471.000000 13 43 1929
690.000000 14 47 2201
626.000000 15 49 2219
626.000000 16 51 2231
633.000000 17 54 2276
358.000000 18 56 2289
260.000000 19 58 2304
650.000000 20 70 2924
586.000000 21 72 2934
709.000000 22 74 3059
821.000000 23 76 3177
714.000000 24 78 3198
676.000000 25 80 3306

```

```
Binder 0  
Arrivals System Server fun obs <Long Delay Time>  
fun obs (bindelem) =  
  if IntInf.toInt(Queue_Delay.last()) >= (!longdelaytime)  
  then 1  
  else 0
```

. 14: Observer Long Delay Time

▀ `longdelaytime`

▼ Declarations

▶ Standard declarations

▶ System

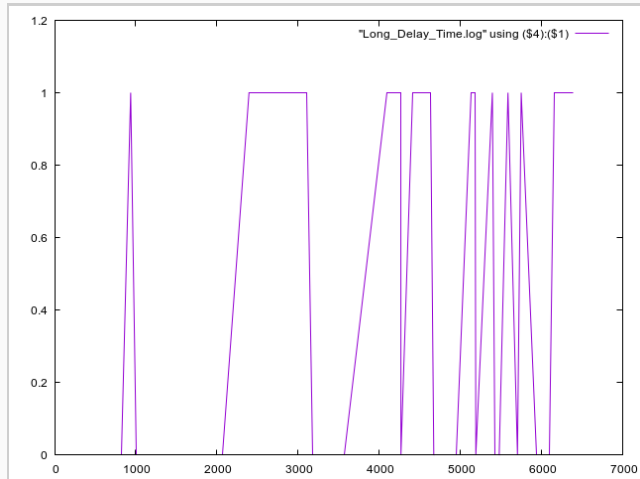
▼ `globref longdelaytime = 200;`

▀ `Monitor`

. 15: `longdelaytime`

```
▼ /home/openmodelica/output/logfiles/Long_Delay_Time.log -
Файл Правка Поиск Вид Документ Справка
#data counter step time
0 1 3 36
0 2 6 282
0 3 9 515
0 4 12 705
0 5 17 826
1 6 21 938
0 7 24 1003
0 8 26 1077
0 9 28 1107
0 10 31 1159
0 11 34 1285
0 12 37 1344
0 13 39 1435
0 14 42 1500
0 15 45 1530
0 16 48 1587
0 17 51 1644
0 18 55 1729
0 19 57 1769
0 20 60 1905
0 21 63 1944
0 22 66 2068
1 23 74 2396
1 24 76 2400
1 25 81 2528
1 26 83 2556
1 27 87 2745
```

. 16: Long_Delay_Time.log



. 17: ,

Tools.

$M|M|1$ CPN